

Lesson 2: Coordinates and Projections

Practical Session: Working with CRSs in QGIS

Objectives

1. Understand how to manipulate Coordinate Reference Systems (CRS) within QGIS.
2. Observe real-time spatial distortion across both vector and raster data layers.
3. Learn the appropriate geoprocessing workflows to reproject data permanently.

Preliminary Setup

1. Launch QGIS and initialize a new, empty project.
2. Locate `ne_10m_admin_0_countries.shp` and `ne_10m_populated_places_simple.shp` vector datasets from your Lesson_1/Data folder and drag them into the map canvas.
3. Import the `HYP_50M_SR_W.tif` (Cross Blended Hypso with Shaded Relief and Water) raster dataset from your Lesson_1/Data folder into QGIS as well.
4. In the Layers Panel, ensure that the populated places and countries vector layers are positioned above the raster dataset for proper visibility.

Step 1: Visualizing Different CRS

QGIS utilizes On-the-Fly projection to display layers with different native coordinate systems simultaneously. Changing the Project CRS alters the rendering of the entire canvas on your screen without modifying the underlying raw data coordinates.

Procedure

1. Observe the bottom right corner of the QGIS interface. By default, the project initializes to EPSG:4326 (WGS 84).
2. Click the EPSG:4326 status bar button to open the Project Properties dialog.
3. In the Filter bar, search for 3857. Select WGS 84 / Pseudo-Mercator (EPSG:3857) and click Apply. Note the severe exaggeration of landmasses at northern/southern latitudes (such as Greenland/Antarctica).
4. Return to the filter bar and search for 3035. Select ETRS89-extended / LAEA Europe (EPSG:3035) and click Apply. The canvas shifts into a circular, globe-like perspective centered on Europe.

Step 2: Measuring Distorted Distances

Because flat maps must compromise on geometric properties, Cartesian (flat-grid) calculations yield vastly different numerical results depending on the active projection.

Procedure

1. Revert your Project CRS to EPSG:4326.
2. Open the Measure Line tool from the Attributes Toolbar (or use the shortcut Ctrl + Shift + M).
3. Set the measurement type to Cartesian and the units to Kilometers.
4. Click once on London, stretch the measurement line to Singapore and click once, and then right-click to terminate the line. Record the calculated distance.
5. Leave the measure dialog open. Change the Project CRS to EPSG:3857 (Web Mercator).
6. Re-measure the distance between London and Singapore.

Note the significant discrepancy in distance caused by the Mercator projection's coordinate stretching away from the equator. By how much do the measurements differ?

Step 3: Permanently Reprojecting Vectors and Rasters

To conduct accurate spatial analysis or geometry calculations, files must natively share the target coordinate system. Changing the visual canvas is insufficient; you must write new data files to disk using geoprocessing tools.

Part 3A: Reprojecting Vectors

1. Navigate to the top menu and open the Processing Toolbox: Go to Processing, then Toolbox.
2. Search for the Reproject Layer tool and double-click to launch it.
3. Input Layer: Select `ne_10m_admin_0_countries`.
4. Target CRS: Select World_Robinson (EPSG:54030).
5. Under Reprojected, click the dropdown menu, select Save to File, and save the output inside Lesson_2/Data/ as `countries_robinson.shp`. Click Run.

Part 3B: Reprojecting Rasters (Warping)

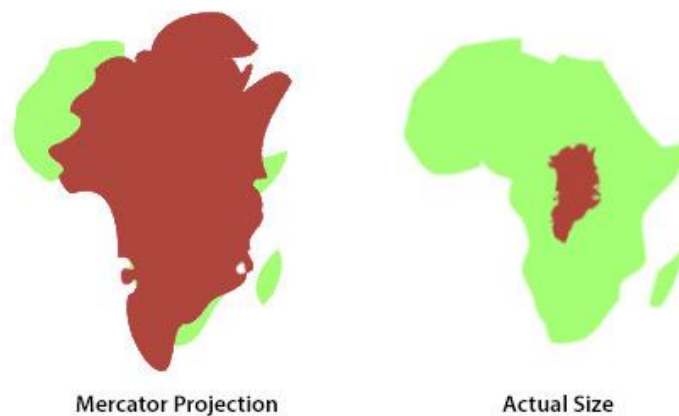
Raster datasets cannot simply shift coordinate points; their pixel grids must be mathematically resampled.

1. In the Processing Toolbox, search for the Warp (reproject) GDAL tool.

2. Input Layer: Select the Natural Earth raster file (Cross Blended Hypso with Shaded Relief and Water).
3. Target CRS: Select World_Robinson (EPSG:54030).
4. Resampling Method: Change this from Nearest Neighbor to Bilinear or Cubic to maintain smooth color gradations.
5. Under Reprojected, select Save to File and name the output `hypso_robinson.tif`. Click Run.

Is Greenland Bigger Than Africa?

Greenland vs Africa



The World Robinson projection (EPSG:54030) does not perfectly preserve area, shape, distance, or direction. Instead, it distorts all four properties slightly to minimize overall visual distortion.

The Problem in Mercator (EPSG:3857): Greenland and Africa appear to be of identical geographic size. In reality, Africa is roughly 14 times larger than Greenland.

By curving the meridians and flattening the poles into lines rather than points, the Robinson projection significantly mitigates high-latitude area exaggeration. Landmasses maintain visually realistic proportions relative to one another, making it the industry standard for thematic world maps.

Summary

1. Match the CRS to the Objective

Geographic Coordinate Systems (GCS): Measured in angular units (Degrees). Use these for data storage, global distribution, and GPS coordinate tracking (such as WGS 84). Do not use them for area or distance measurements.

Projected Coordinate Systems (PCS): Measured in linear units (Meters or Feet). Use these for accurate localized mapping, buffer generation, and spatial calculations (such as UTM or regional state planes).

2. Verify Project Settings Continuously

Always check the active EPSG code in the bottom-right corner before finalizing layout components like scale bars. A mismatched project projection will render scale calculations invalid.

3. Changing Project CRS vs. Permanent Reprojection

On-the-Fly projection only alters visual representation. Permanent alterations to coordinate systems require saving a new file via the Processing Toolbox using Reproject Layer or Warp.

Homework: Visualizing Coordinate Grids (GCS vs. PCS)

Objective

Create a comparative map layout demonstrating how coordinate values change from angular degrees (GCS) to linear meters (PCS) across the same geographic region.

Layout Setup

1. Open QGIS and load the `ne_10m_admin_0_countries.shp` layer.
2. Center the map on Germany at a readable scale.
3. Navigate to Project, then New Print Layout. Name it Homework_2.
4. Set the page orientation to Landscape (if not set already).

Configure Map 1 (Geographic Coordinate System)

1. Use the Add Map tool to draw a box on the left half of the page.
2. In the Item Properties panel on the right, locate the CRS dropdown and explicitly set it to EPSG:4326 - WGS 84.

3. Scroll down to Grids, click the green plus icon (+), select Grid 1, and click Modify Grid...
4. Set both X and Y Interval to 10.0 (this value represents 10 degrees).
5. Change the Frame style to Interior ticks.
6. Check the Draw Coordinates box. Change the Format to Decimal with suffix.
7. In the Item Properties panel, under Layers section, check the Lock Styles and Lock styles for layer boxes to lock the map.

Configure Map 2 (Projected Coordinate System)

1. Use the Add Map tool to draw a second box on the right half of the page.
2. In the Item Properties panel, set the CRS to EPSG:3857 - WGS 84 / Pseudo-Mercator.
3. Scroll down to Grids, click the green plus icon (+), select Grid 1, and click Modify Grid...
4. Set both X and Y Interval to 1000000.0 (this value represents 1,000,000 m).
5. Change the Frame style to Interior ticks.
6. Check the Draw Coordinates box. Change the Format to Decimal.

Finalize Layout

1. Use the Add Label tool to add a suitable title.
2. Add labels below each map identifying the projection used, such as "Left: GCS WGS 84 (EPSG:4326)" and "Right: PCS Web Mercator (EPSG:3857)".
3. Export the final page as a PDF via Layout, then Export as PDF...

Submission Deliverables

Submit the exported single-page PDF map. Beneath the maps, include a very short text block responding to the following questions:

Question 1: What are the coordinate units displayed on the left map versus the right map?

Question 2: Look closely at the coordinate values for Berlin on both maps. Approximate the coordinate readings for Berlin in both systems based on your grid lines.